ZUKUNFT LAND

# Rheingau

Full data report

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Hesse / Rhineland-Palatinate

**NUTS-3 regions:** DE71D, DEB11, DEB17, DEB1A, DEB1D, DEB3J

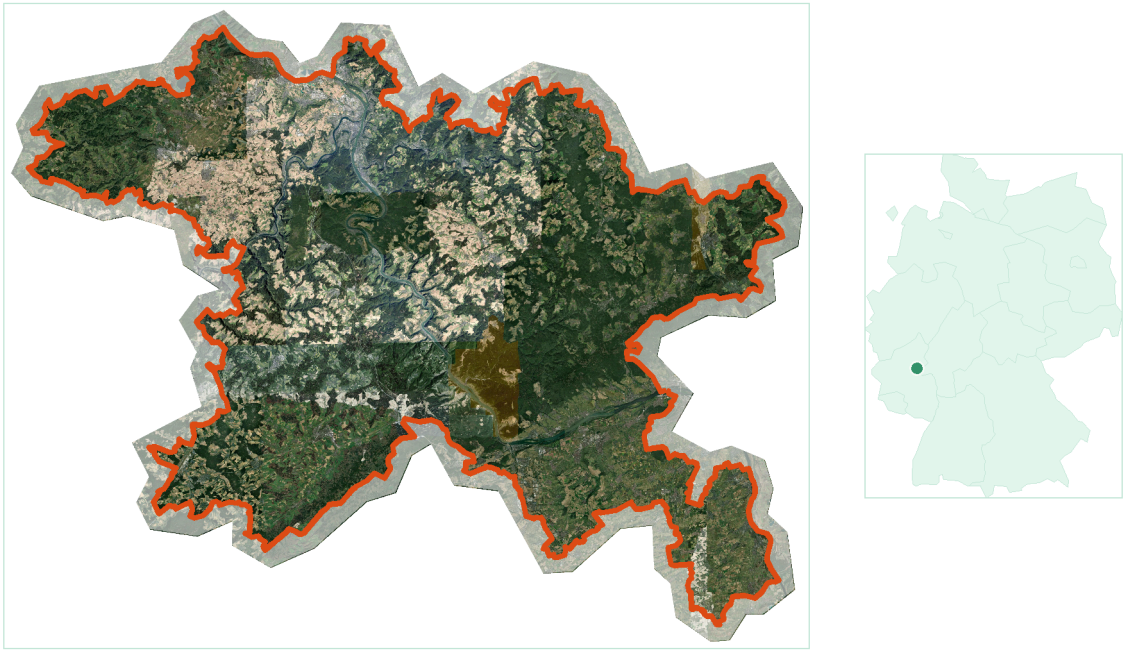


Figure 1: The Rheingau Living Lab: study region (left) and its location within Germany (right). Basemap: Tiles © Esri - Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

# Agriculture

## Key figures

| CROPLAND AREA | NUMBER OF FARMS | AVERAGE FARM SIZE | SHARE OF ORGANIC FARMING |
|---|---|---|---|
| 104,899 ha | 3,426 count | 45.5 ha | 7.5 % |

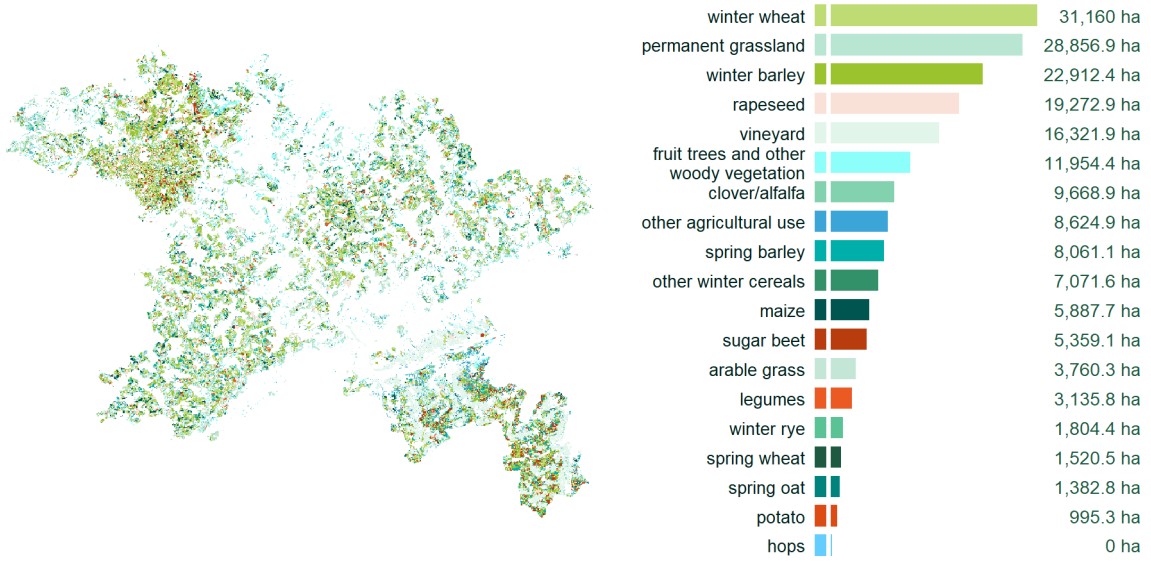



Figure 2: Map showing Agriculture in the Rheingau Living Lab (Data: Crop types (DLR, 2024))

## Overview

The Rheingau region is defined by viticulture and is known for the white grape variety Riesling as well as Pinot noir among the red varieties. For the most part, it is small family-run businesses that cultivate the vineyards, alongside historic institutions such as Eberbach Abbey.

## Challenges

(German) wine is in a structural market crisis: bulk wine prices have fallen below production costs, domestic demand for alcoholic drinks keeps declining, and trade barriers make exports harder. Steep-slope viticulture is highly labour-intensive and qualified seasonal workers are scarce, so rising wages, energy costs and regulatory burden bite hardest exactly where the landscape value is greatest. Small, fragmented holdings limit mechanisation and make succession increasingly uncertain.

# Climate

## Key figures

| GROWING DEGREE DAYS, ANNUAL SUM (BASE 5 °C) | MEAN ANNUAL TEMPERATURE | ANNUAL PRECIPITATION | PRECIPITATION OF WARMEST QUARTER |
|---|---|---|---|
| **1,885** °C·d<br>+1,119 °C·d by 2071-2100 | **9.1** °C<br>+3.9 °C by 2071-2100 | **722** mm<br>+1.5 % by 2071-2100 | **199** mm<br>-10.4 % by 2071-2100 |

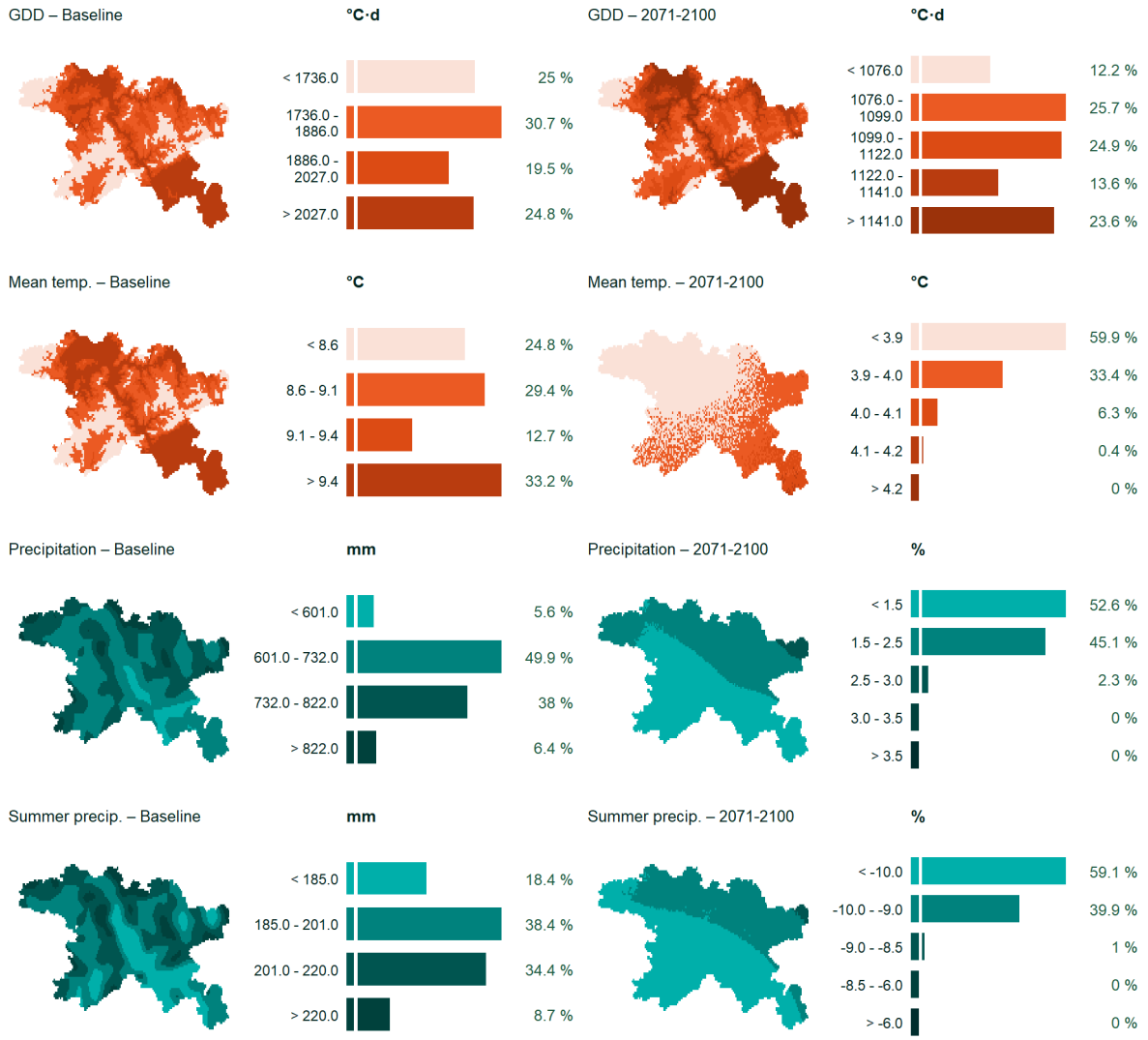



**Figure 3: Map showing Climate in the Rheingau Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)**

Heat accumulated above 5 °C over the year - a measure of how much growing season a crop gets.
Mean annual air temperature, averaged over the reference period.
Total precipitation summed over the year.
Precipitation falling in the warmest three-month quarter - relevant to summer drought stress.

Rheingau

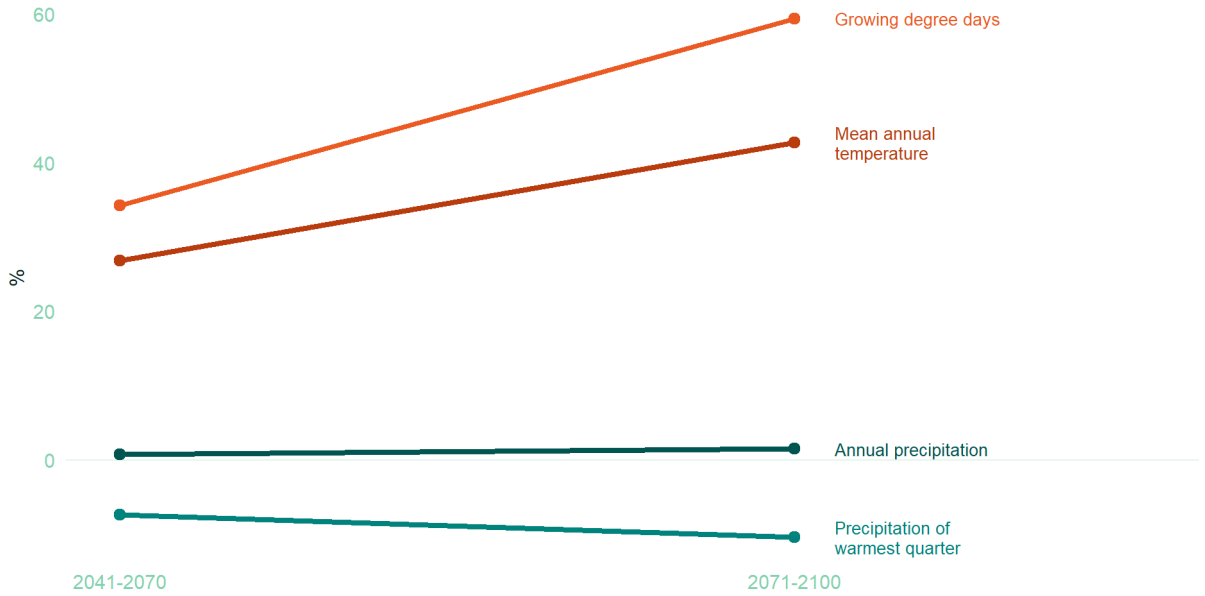



**Figure 4: Chart showing Climate in the Rheingau Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)**

## Overview

Sheltered from northerly winds by the Taunus ridge and favoured by the Rhine, which stores heat and reflects light onto the south-facing slopes. The Rheingau is one of the mildest parts of Germany with low precipitation.

## Challenges

Warming leads to an earlier budburst and ripening, higher sugar and alcohol levels and falling acidity threaten the filigree style the region is known for. Earlier budburst also raises exposure to late frost, while more hot days bring sunburn on the grapes. The increase of precipitation extremes (Drought and heavy rain) is leading in one year to drought stress and in the other to high disease pressure by fungi.

# Soil

## Key figures

| NITROGEN SURPLUS | PHOSPHORUS SURPLUS | GROUNDWATER ABSTRACTION (NON-PUBLIC SUPPLY) |
|---|---|---|
| **No data available** kg N/ha UAA | **No data available** kg P/ha UAA | **9,740** 1000 m3 |

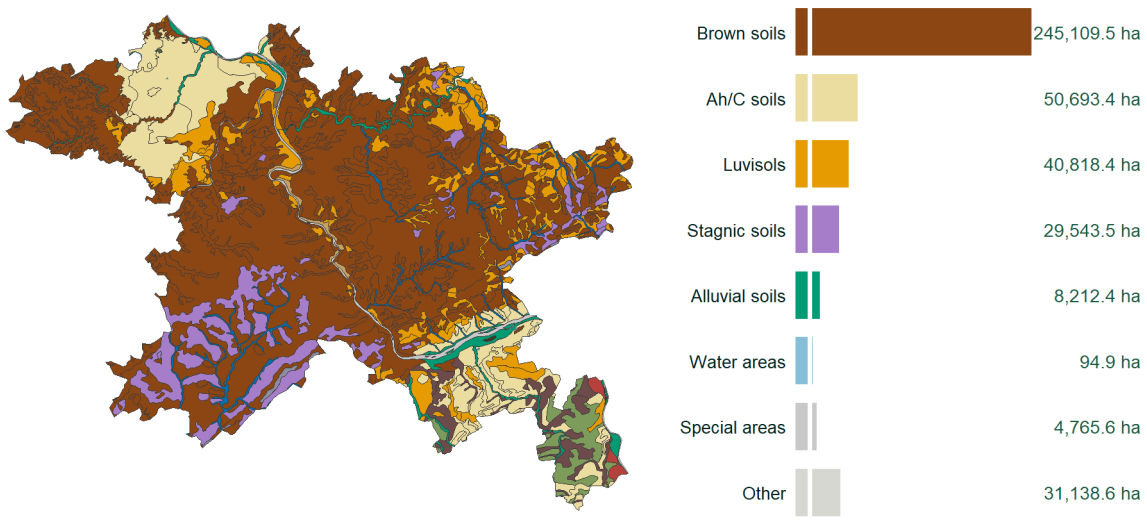



**Figure 5: Map showing Soil in the Rheingau Living Lab (Data: Soil overview map (BUEK250))**
Legend shows the dominant semantic soil groups for this Living Lab; raw BUEK IDs stay in the data only as provenance.

## Overview

Soils change over very short distances. Deep, fertile loess and loess loam cover the gentler lower slopes and the plain near the Rhine, Taunus quartzite dominates the higher sites of the central and eastern Rheingau and the Lorch area, and phyllite and Hunsrück slate carry the vineyards of Rüdesheim, Assmannshausen, Kiedrich, Martinsthal and Rauenthal. The stony slope soils are shallow with low water storage capacity, while the loess soils of the plain are among the most productive arable soils in the region.

## Challenges

Heavy rainfall causes soil erosion on the steep slopes, particularly where rows run downslope and the ground is kept bare. Warmer soils mineralise organic matter faster, so humus is lost just where it matters most for water retention and nutrient supply. Machinery traffic compacts the soil, and nutrient leaching into groundwater remains a concern in the permanent-crop areas.

# Socio-economic

## Key figures

| TOTAL POPULATION | GDP PER CAPITA | UNEMPLOYMENT RATE | DISPOSABLE HOUSEHOLD INCOME PER CAPITA |
|---|---|---|---|
| 954,163 persons | 48,126.5 EUR | 4.92 % | 29,344 EUR |

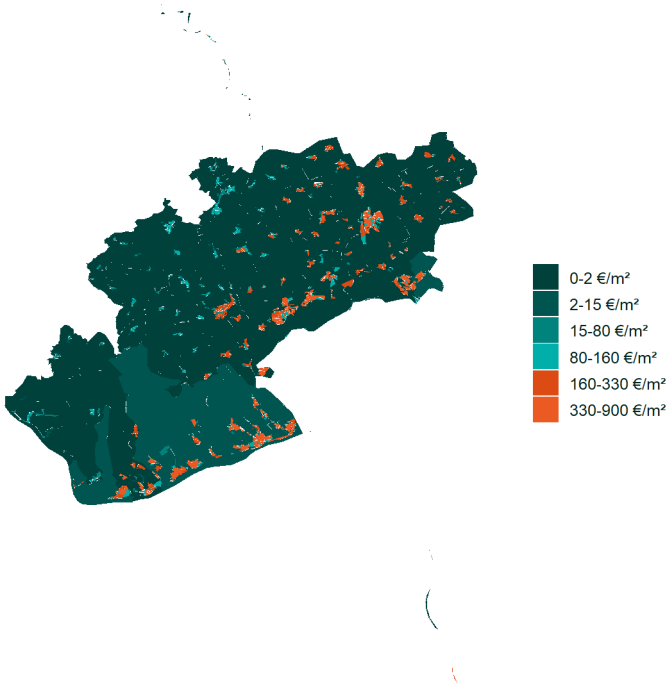



Figure 6: Map showing Socio-economic in the Rheingau Living Lab (Data: Standard land values (BORIS))

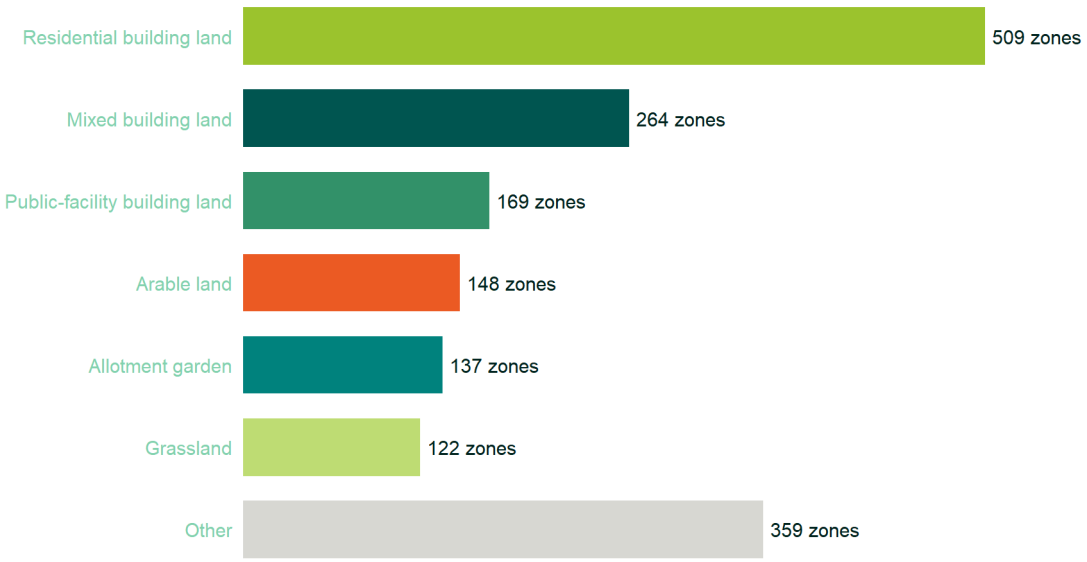



**Figure 7: Chart showing Socio-economic in the Rheingau Living Lab (Data: Standard land values (BORIS))**

## Overview

The economy is dominated by small and medium-sized firms, including high-tech in information and communication technology, sensor and control engineering, medical, surface and environmental technology. Commuters, mainly towards Wiesbaden, Mainz and Frankfurt. Wine and tourism are the signature sectors, strong day-visitor flows, drawn by Riesling, Rhine romanticism and events such as the Rheingau Musik Festival.

## Challenges

Whole villages depend economically, socially and culturally on wine: if estates disappear, the result is not only scrubbed-over slopes but empty villages, closed wine taverns and a weakened tourism offer. Falling wine prices and rising costs squeeze margins, and businesses compete for skilled labour with the far larger Rhine-Main job market, which also drives up land and housing costs.

# Landscape

## Key figures

| FOREST AREA | NATURA 2000 AREA | NATURE RESERVES AREA | SHARE OF SEALED SURFACE |
|---|---|---|---|
| 159,606 ha | 72,770 ha | 13,368.6 ha | 15.4 % |

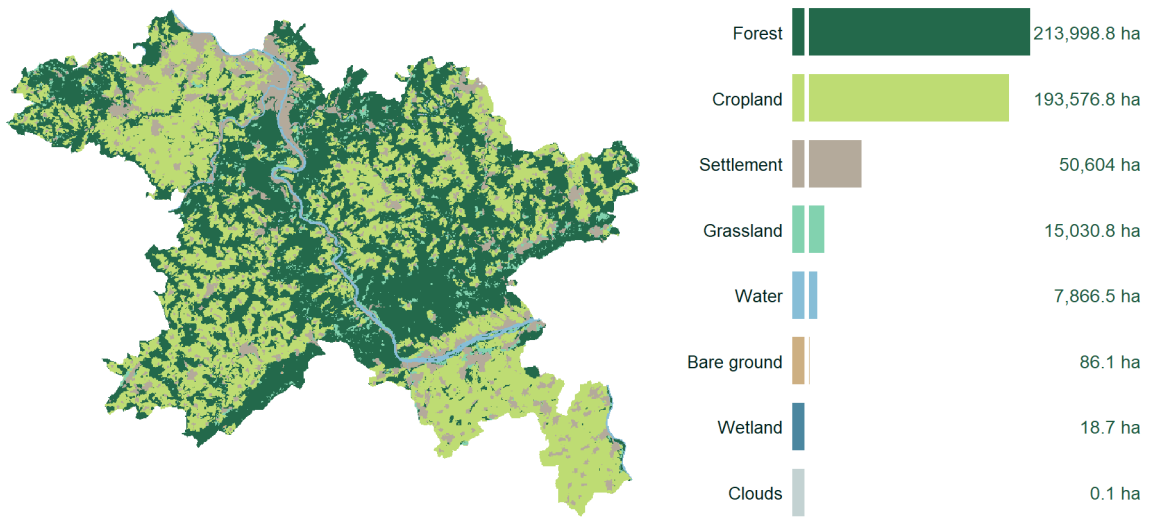



**Figure 8: Map showing Landscape in the Rheingau Living Lab (Data: Land cover (Esri/Impact Observatory, 2024))**

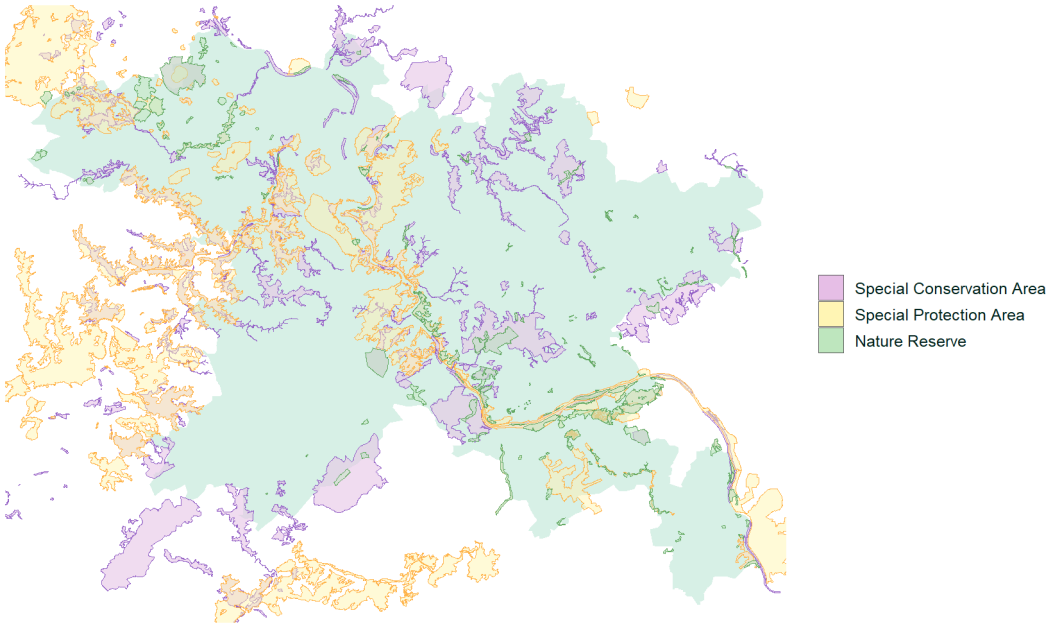



**Figure 9: Map showing Protected Areas in the Rheingau Living Lab (Data: Protected areas (BfN, 2019/2023))**

Conservation sites intersecting this Living Lab region. Full polygon boundaries are shown, so sites extend beyond the region outline and designations may overlap. Source: BfN (FFH and bird sanctuaries 2019, nature reserves 2023) - not suitable for planning purposes.

## Overview

An east—west band between the Rhine and the Taunus, with south-facing slopes and river terraces carrying steep-slope and terraced vineyards below the wooded Rheingaugebirge. Forest cover is well above the Hessian average. The lower Rheingau from Rüdesheim to Lorchhausen belongs to the UNESCO World Heritage site 'Upper Middle Rhine Valley', with the Limes as a second World Heritage site in the district. The Rhein-Taunus Nature Park, the Wisper, Aar and Wörsbach valleys, monasteries such as Kloster Eberbach and historic wine villages complete a densely layered cultural landscape.

## Challenges

Steep slopes and terraces can only be worked with high manual effort; where cultivation is abandoned, scrub encroachment erases both the terraced cultural landscape and the dry, warm habitats it carries. Heritage and scenic protection set narrow limits for new infrastructure, while land is simultaneously sought for renewable energy and settlement growth. Small, fragmented parcels make land consolidation difficult, and drought stress is increasingly visible in the forests.